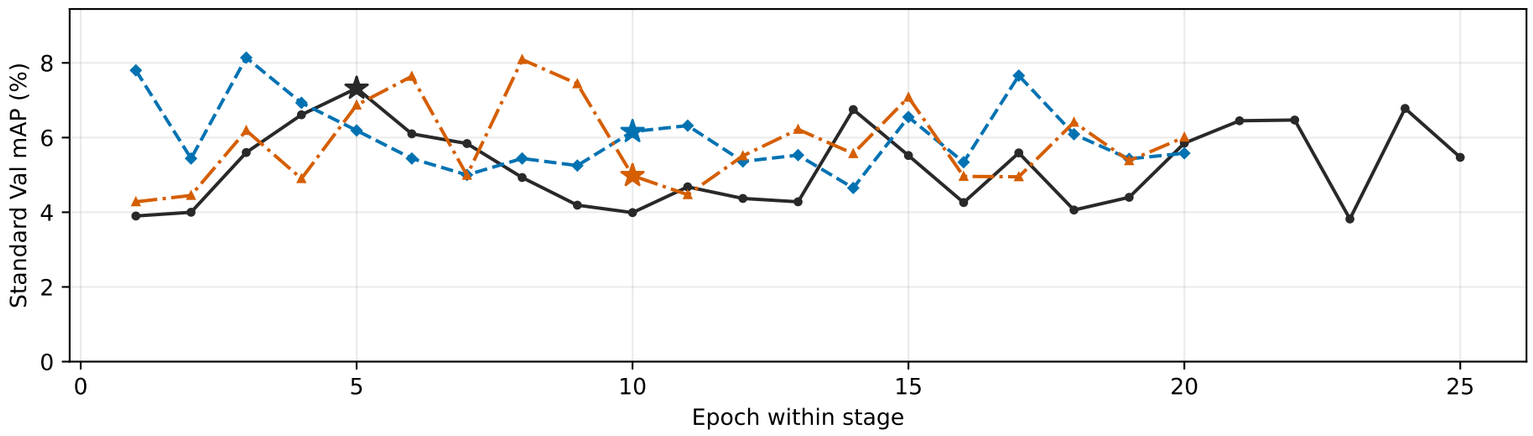
Moment-DETR | GMR → original GEQ vs. joint auxiliary GEQ

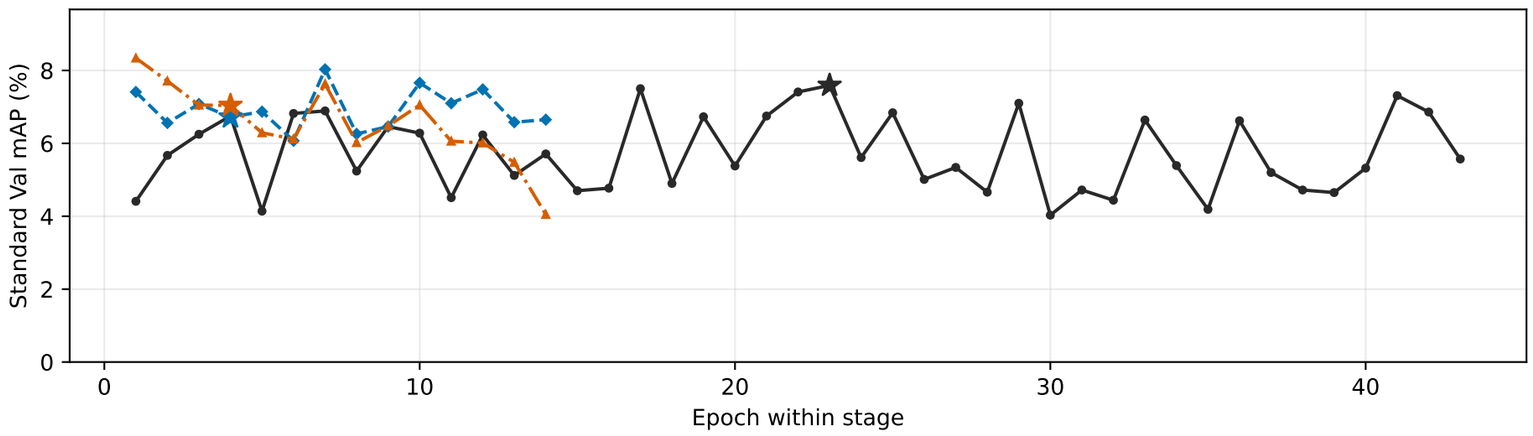
E1: Balanced BCE



Stage	State	Last / picked mAP
GMR	done	5.47 / 7.32 picked epoch 5
GEQ original	done	5.58 / 6.16 picked epoch 10
GEQ + aux	done	6.01 / 4.98 picked epoch 10

GMR Val clip AUPRC 0.124 | GT/background gap 0.144

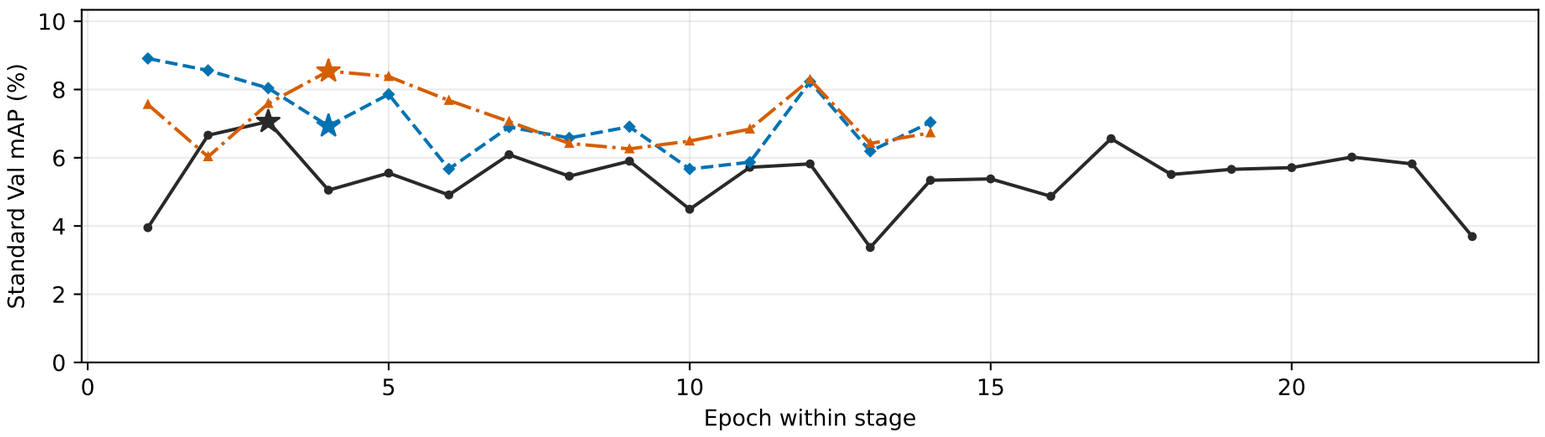
E2: Balanced Focal



Stage	State	Last / picked mAP
GMR	done	5.57 / 7.59 picked epoch 23
GEQ original	done	6.65 / 6.72 picked epoch 4
GEQ + aux	done	4.05 / 7.04 picked epoch 4

GMR Val clip AUPRC 0.140 | GT/background gap 0.125

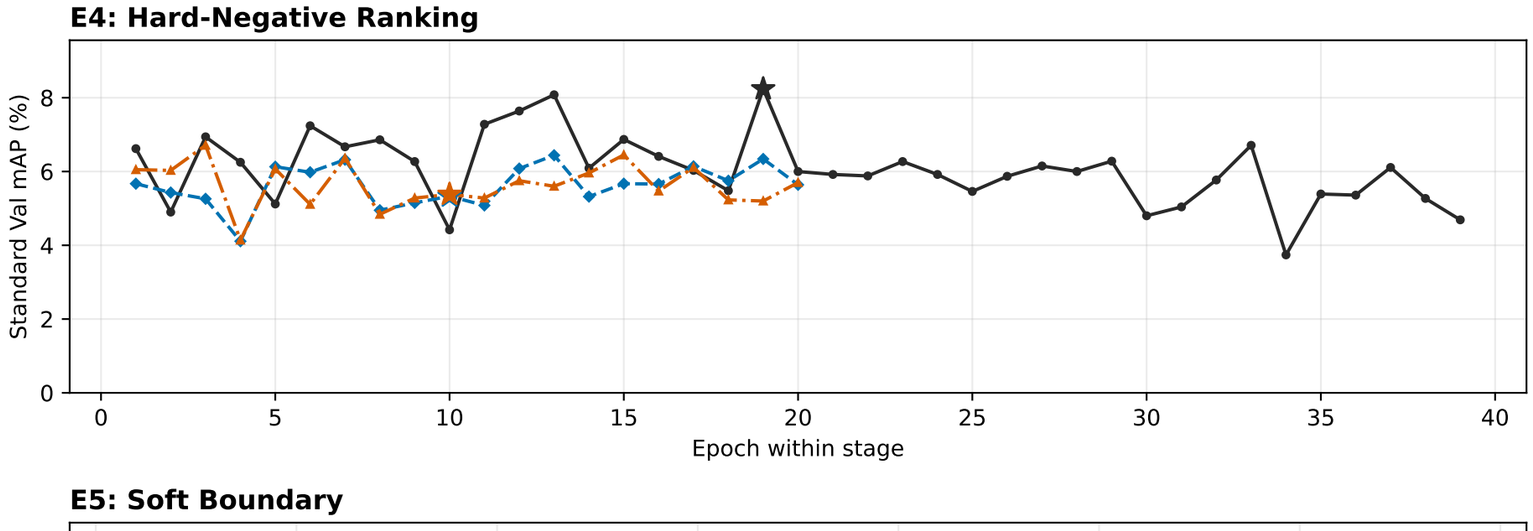
E3: Balanced BCE + Dice



Stage	State	Last / picked mAP
GMR	done	3.69 / 7.06 picked epoch 3
GEQ original	done	7.04 / 6.93 picked epoch 4
GEQ + aux	done	6.73 / 8.54 picked epoch 4

GMR Val clip AUPRC 0.098 | GT/background gap 0.081

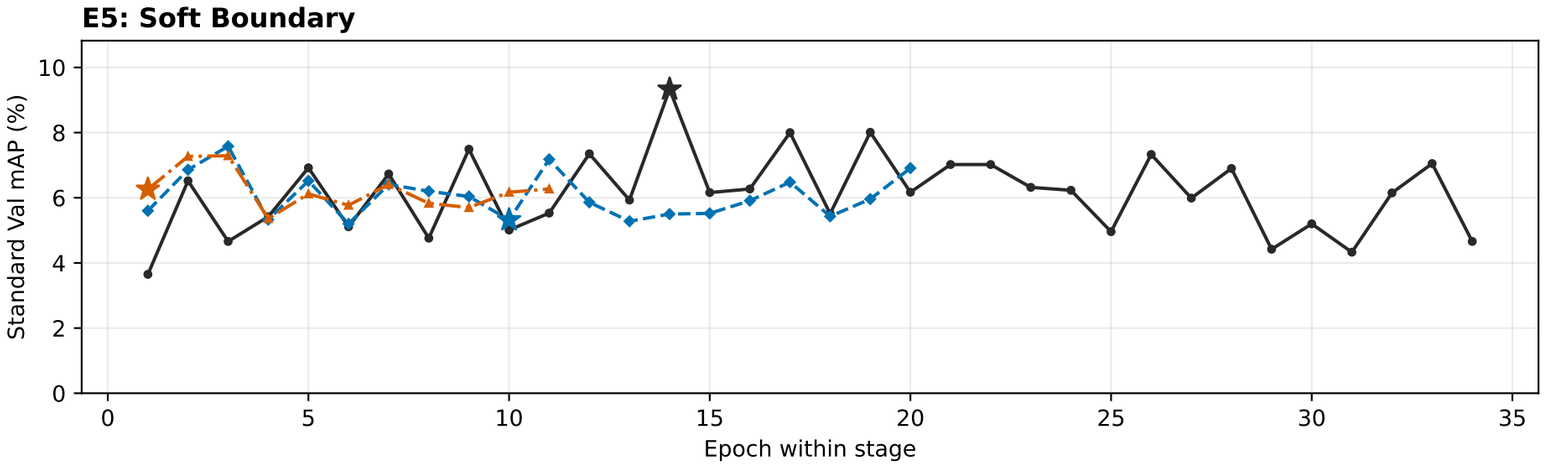
E4: Hard-Negative Ranking



Stage	State	Last / picked mAP
GMR	done	4.69 / 8.24 picked epoch 19
GEQ original	done	5.64 / 5.33 picked epoch 10
GEQ + aux	done	5.70 / 5.38 picked epoch 10

GMR Val clip AUPRC 0.126 | GT/background gap 0.172

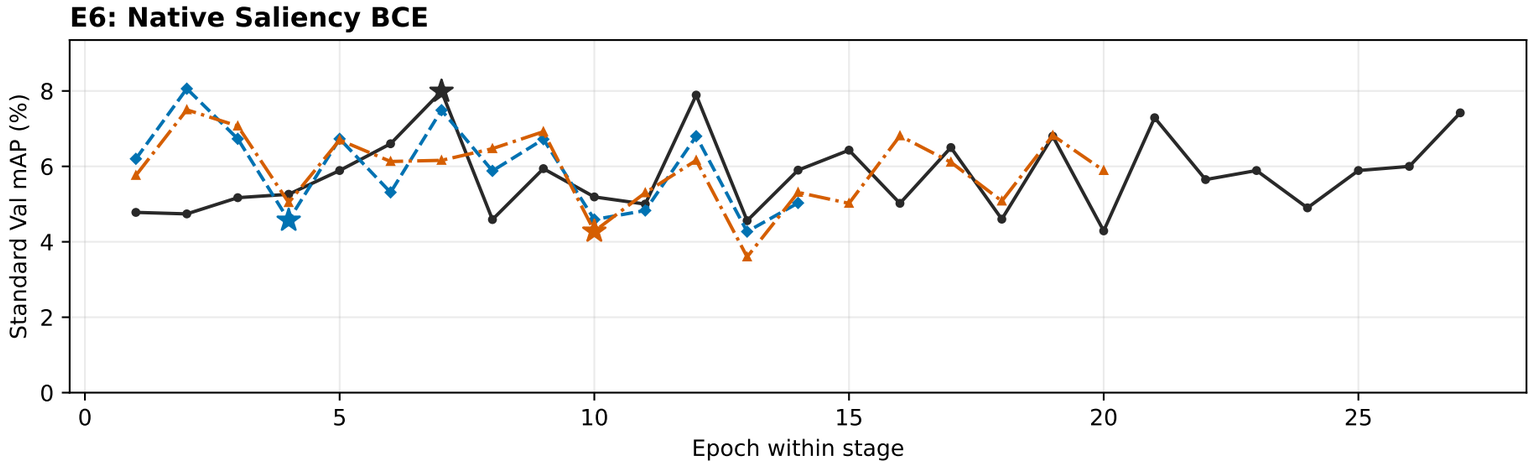
E5: Soft Boundary



Stage	State	Last / picked mAP
GMR	done	4.66 / 9.33 picked epoch 14
GEQ original	done	6.91 / 5.32 picked epoch 10
GEQ + aux	done	6.27 / 6.26 picked epoch 1

GMR Val clip AUPRC 0.156 | GT/background gap 0.123

E6: Native Saliency BCE



Stage	State	Last / picked mAP
GMR	done	7.42 / 7.99 picked epoch 7
GEQ original	done	5.03 / 4.57 picked epoch 4
GEQ + aux	done	5.89 / 4.28 picked epoch 10

GMR Val clip AUPRC 0.088 | GT/background gap 0.051

GMR selected by Val mAP; GEQ by Val mean G1 (.4/.5/.6). Test excluded. 2026-11-03T20:57:15